Auteur: Yoren Collier
Studierichting: Informatica
Oefeningenreeks 3B nr. 3

$$f(x) = \frac{1}{\sqrt{x}} = x^{\frac{-1}{2}}$$
$$\frac{df}{dx}(x) = \frac{-1}{2}\,(x)^{\frac{-3}{2}}$$
$$\frac{d^2f}{dx^2}(x) = \frac{-1}{2} \cdot \frac{-3}{2}\,(x)^{\frac{-5}{2}}$$
$$\frac{d^3f}{dx^3}(x) = \frac{-1}{2} \cdot \frac{-3}{2} \cdot \frac{-5}{2}\,(x)^{\frac{-7}{2}}$$
$$\frac{d^4f}{dx^4}(x) = \frac{-1}{2} \cdot \frac{-3}{2} \cdot \frac{-5}{2} \cdot \frac{-7}{2}\,(x)^{\frac{-9}{2}} = \frac{+105}{16\sqrt{x^9}} = \frac{105}{16x^4\sqrt{x}} = \frac{105\sqrt{x}}{16x^5}$$

References